

TRIBHUVAN UNIVERSITY

Institute of Science and Technology

NET CENTRIC COMPUTING

Five Critical Model Question Sets for Examination Preparation

For B.Sc. CSIT (CSC367) and BIT (BIT351)

[VVI] Very Very Important questions are highlighted in light yellow.

Prepared by

Mr Subash Pariyar

Exam-oriented, unit-balanced, and critically selected practice sets

MODEL QUESTION SET 1

Subject: NET Centric Computing	Full Marks: 60	Time: 3 Hours
Programmes: B.Sc. CSIT (CSC367) and BIT (BIT351)	Pass Marks: 24	Prepared by: Mr Subash Pariyar

SECTION A: LONG ANSWER QUESTIONS

Attempt any TWO questions. [2 x 10 = 20]

- [VVI] 1. Critically differentiate between an abstract class, interface, and sealed class in C#. Explain when each should be used. Write a program that uses an abstract class with an abstract method and a sealed class to prevent further inheritance. [4 + 6]**
- [VVI] 2. Explain controllers, action methods, and Action Result types in ASP.NET Core MVC. Design an API controller for Product(ProductId, Name, Price, Category) and implement GET, POST, PUT, and DELETE operations that return appropriate JSON responses. [4 + 6]**
- [VVI] 3. Explain the roles of Connection, Command, DataReader, and DataAdapter in ADO.NET. Given BOOK(Id, Name, Author, Price), write an ADO.NET program to retrieve books having a price greater than 500 and explain how parameterized queries protect the application from SQL injection. [4 + 6]**

SECTION B: SHORT ANSWER QUESTIONS

Attempt any EIGHT questions. [8 x 5 = 40]

- [VVI] 4. Explain MSIL and CLR and describe their roles in the execution of a .NET application. [5]**
- [VVI] 5. Explain the MVC pattern and justify why it is suitable for developing dynamic web applications. [5]**
6. Differentiate between model binding and model validation. Explain validation using data annotations. [5]
- [VVI] 7. How is state maintained over stateless HTTP? Explain Session State and cookies. [5]**
8. Write a jQuery program to validate username, password, and a ten-digit phone number. [5]
- [VVI] 9. Explain Dependency Injection and the IoC container. Differentiate between transient, scoped, and singleton services. [5]**
10. Differentiate between roles, claims, and policies in ASP.NET Core authorization. [5]
11. Explain Kestrel and its relationship with IIS or Nginx. [5]
12. Discuss Docker and containerization in ASP.NET Core application deployment. [5]

MODEL QUESTION SET 2

Subject: NET Centric Computing	Full Marks: 60	Time: 3 Hours
Programmes: B.Sc. CSIT (CSC367) and BIT (BIT351)	Pass Marks: 24	Prepared by: Mr Subash Pariyar

SECTION A: LONG ANSWER QUESTIONS

Attempt any TWO questions. [2 x 10 = 20]

[VVI] 1. Explain the principle of asynchronous programming. Why is asynchronous programming important in web applications? Write a C# program using async and await and explain its execution flow. [4 + 6]

[VVI] 2. Define URL routing and explain its major features. Describe how conventional routing and attribute routing map an HTTP request to a controller action. Illustrate with suitable examples. [5 + 5]

[VVI] 3. Explain Entity Framework Core and the role of an Object-Relational Mapper. Given Employee(EmployeeId, Name, Department, Salary), write EF Core code to insert an employee, retrieve employees with salary greater than 50,000, update an employee salary, and delete an employee. [4 + 6]

SECTION B: SHORT ANSWER QUESTIONS

Attempt any EIGHT questions. [8 x 5 = 40]

[VVI] 4. Differentiate between .NET, .NET Core, and Mono. [5]

5. What are named and positional attribute parameters? Explain with an example. [5]

[VVI] 6. Explain Razor syntax and show how C# code is embedded in an HTML view. [5]

7. What are Tag Helpers? Explain any three commonly used Tag Helpers. [5]

8. Create a controlled React form that accepts student name, email, and course. [5]

[VVI] 9. What is ASP.NET Core Identity? Explain the purpose of authentication. [5]

10. Define Cross-Site Scripting and explain its preventive measures. [5]

11. Differentiate between TempData, ViewData, and Session State. [5]

12. Explain the steps required to build, test, publish, and deploy a .NET Core application. [5]

MODEL QUESTION SET 3

Subject: NET Centric Computing	Full Marks: 60	Time: 3 Hours
Programmes: B.Sc. CSIT (CSC367) and BIT (BIT351)	Pass Marks: 24	Prepared by: Mr Subash Pariyar

SECTION A: LONG ANSWER QUESTIONS

Attempt any TWO questions. [2 x 10 = 20]

[VVI] 1. Differentiate between generic and non-generic collections. Design a generic class containing a generic member variable, generic property, generic constructor, and generic method. Explain how generics improve type safety and code reusability. [4 + 6]

[VVI] 2. Explain HTTP request and response message formats. Trace the complete processing of an HTTP request through routing, controller, model, view, and response in an ASP.NET Core MVC application. [4 + 6]

[VVI] 3. What is SQL injection? Analyze the programming errors that make an application vulnerable to SQL injection. Write vulnerable and secure versions of a database query and explain the prevention mechanisms. [4 + 6]

SECTION B: SHORT ANSWER QUESTIONS

Attempt any EIGHT questions. [8 x 5 = 40]

4. Explain delegates and events and describe their relationship with an example. [5]

5. Differentiate between method hiding and method overriding. [5]

[VVI] 6. Explain data models and DbContext in Entity Framework Core. [5]

7. What is a Single-Page Application? State its advantages and limitations. [5]

8. Explain hidden fields and query strings as client-side state-management techniques. [5]

9. Why is client-side form validation necessary? Give an example. [5]

[VVI] 10. Explain Cross-Site Request Forgery and describe the role of anti-forgery tokens. [5]

11. What is the ASP.NET Core Module? Explain how it supports hosting with IIS. [5]

12. Explain the procedure for publishing an ASP.NET Core application to Microsoft Azure. [5]

MODEL QUESTION SET 4

Subject: NET Centric Computing	Full Marks: 60	Time: 3 Hours
Programmes: B.Sc. CSIT (CSC367) and BIT (BIT351)	Pass Marks: 24	Prepared by: Mr Subash Pariyar

SECTION A: LONG ANSWER QUESTIONS

Attempt any TWO questions. [2 x 10 = 20]

- [VVI] 1. Explain inheritance and polymorphism in C#. Discuss the use of the base keyword, method overriding, and dynamic method dispatch. Write a program using EMPLOYEE as a base class and ENGINEER and DOCTOR as derived classes. [4 + 6]**
- [VVI] 2. Describe the ASP.NET Core architecture. Compare it with the traditional ASP.NET Framework architecture in terms of platform support, dependency management, hosting, middleware, performance, and deployment. [5 + 5]**
- [VVI] 3. Compare ADO.NET and Entity Framework Core. Given FACULTY(CourseId, CourseName, NumberOfSemesters, Fee), write code to insert records and retrieve the fee of the course whose name is CSIT. Discuss whether ADO.NET or EF Core is more suitable for the operation. [4 + 6]**

SECTION B: SHORT ANSWER QUESTIONS

Attempt any EIGHT questions. [8 x 5 = 40]

4. Differentiate between a structure and an enumeration. [5]
- [VVI] 5. Explain user-defined exceptions with a suitable C# example. [5]**
6. What are partial and sealed classes? Explain their practical uses. [5]
- [VVI] 7. How is HTML rendered using MVC views? Explain the role of Razor. [5]**
8. Explain caching as a server-side state-management strategy. [5]
9. Compare Angular and React for building a Single-Page Application. [5]
10. Define open redirect attack and explain its prevention. [5]
- [VVI] 11. How are controllers and action methods secured using authorization attributes? [5]**
12. Compare IIS, Nginx, Apache, and Kestrel as hosting options. [5]

MODEL QUESTION SET 5

Subject: NET Centric Computing	Full Marks: 60	Time: 3 Hours
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SECTION A: LONG ANSWER QUESTIONS

Attempt any TWO questions. [2 x 10 = 20]

[VVI] 1. Differentiate between errors and exceptions. Explain the importance of exception handling. Write a program that accepts marks for five subjects and throws a custom exception when a mark is negative or exceeds 100. [3 + 7]

[VVI] 2. Explain Dependency Injection and the Inversion of Control container. Design a service and controller demonstrating constructor injection. Discuss transient, scoped, and singleton service lifetimes and identify appropriate use cases for each. [4 + 6]

[VVI] 3. Explain authentication and authorization in ASP.NET Core. Describe the process of configuring ASP.NET Core Identity and securing controllers through roles, claims, and policies. [4 + 6]

SECTION B: SHORT ANSWER QUESTIONS

Attempt any EIGHT questions. [8 x 5 = 40]

4. Define indexers and distinguish them from ordinary properties. [5]

[VVI] 5. Explain the compilation and execution process of a .NET application using CLI, MSIL, CLR, JIT, and native code. [5]

[VVI] 6. Explain Action Result types in ASP.NET Core MVC with suitable examples. [5]

7. What is JSON? Explain its use in ASP.NET Core Web API applications. [5]

8. Write an Entity Framework Core query to update the price of books written by John to 1000. [5]

[VVI] 9. Compare cookies, Session State, TempData, query strings, and hidden fields. [5]

10. Explain the causes, consequences, and prevention of Cross-Site Scripting attacks. [5]

11. Explain the importance of Single-Page Applications and REST API integration. [5]

12. Describe the uses of Docker and explain the basic containerization process for an ASP.NET Core application. [5]

FINAL VVI REVISION CHECKLIST

Prepare these areas first before attempting the five complete sets.

- 1. Controllers, actions, Action Result types, and Web API CRUD operations**
- 2. Dependency Injection, IoC container, and service lifetimes**
- 3. ADO.NET classes and database operations**
- 4. Entity Framework Core, ORM, data model, DbContext, and CRUD**
- 5. SQL injection attack, vulnerable code, and prevention**
- 6. MVC pattern, HTTP flow, and ASP.NET Core architecture**
- 7. Collections, generics, generic class, and type safety**
- 8. Exception handling and user-defined exceptions**
- 9. State management: Session, TempData, cookies, query strings, hidden fields, and cache**
- 10. Authentication, authorization, roles, claims, policies, and ASP.NET Core Identity**
- 11. Razor syntax, views, Tag Helpers, model binding, and validation**
- 12. React or jQuery form creation and validation**
- 13. Kestrel, IIS, Nginx, Docker, Azure, and deployment**
- 14. CLI, MSIL, CLR, JIT, and .NET execution process**
- 15. SPA, Angular, React, and REST API integration**

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